B.M.S. College of Engineering

## Bio-inspired Algorithms | 23DS7BSBIO

# Concrete Compressive Strength Prediction Using Differential Evolution–Optimized Support Vector Regression

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AY 2026–27

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# Introduction to Optimization Problems

## Objective and search space

Optimization selects parameters that minimize an objective.

Here: minimize five-fold development RMSE of an RBF-SVR.

Decision variables: $\log_{10}(C)$, $\log_{10}(\gamma)$, $\log_{10}(\varepsilon)$.

Constraints: fixed parameter bounds and 300 objective calls.

The objective is evaluated by fitting models; derivatives are unavailable.

Holdout performance evaluates fixed selections after tuning.

# What are Nature-Inspired Algorithms?

## Nature-inspired search

Search methods motivated by biological, collective or physical processes.

They explore candidate solutions and use measured fitness to guide updates.

Genetic Algorithm: selection, crossover and mutation.

Differential Evolution: mutation from population-vector differences.

Particle Swarm Optimization: individual and social search information.

No gradient is required; performance still depends on task and controls.

# Classification of Nature-Inspired Algorithms

## Algorithm families

Evolutionary methods: Genetic Algorithms and Differential Evolution.

Swarm methods: Particle Swarm Optimization and Ant Colony Optimization.

Physical-process methods: Simulated Annealing.

This project implements DE/rand/1/bin for continuous parameters.

Random Search is the equally budgeted tuning comparator.

The other algorithms are background methods, not measured baselines.

# Problem Statement

## Prediction task

Predict concrete compressive strength in MPa from ingredients and age.

SVR accuracy depends on regularization C, kernel γ and tolerance ε.

Question: does DE improve tuning under an equal evaluation budget?

Compare default SVR, Random Search–SVR and DE–SVR.

Keep data splits, preprocessing, search bounds and budgets fixed.

Report all predefined seeds and the limits of the comparison.

# Dataset and Preprocessing

## Dataset and evaluation split

UCI Concrete Compressive Strength: 1,030 rows and eight inputs.

Seven ingredient quantities (kg/m³) and curing age (days).

Target: compressive strength (MPa); no missing values.

996 unique input tuples; identical inputs stay in the same group.

Development: 826 rows / 796 groups. Holdout: 204 rows / 200 groups.

Five group-disjoint CV folds; StandardScaler fitted within each fold.

Use original features; retain every observation.

# Implementation and Results

## Pipeline

Raw-data audit and saved group splits

StandardScaler → RBF-SVR

Development CV evaluates candidates

Refit selected parameters on development

Evaluate on fixed holdout

Save histories, predictions and provenance

## DE configuration

Population: 15; F = 0.8; CR = 0.9

300 calls including initialization

19 trial generations after initialization

Seeds: 11, 22, 33, 44, 55

$\log_{10}$ C: [−2, 4]; $\log_{10}$ γ: [−5, 1]

$\log_{10}$ ε: [−3, 1]

# Differential Evolution: Explicit Operators

## DE/rand/1/bin

1. Initialize 15 candidates uniformly within the log bounds.
2. Select three distinct donors, all excluding the current target.
3. Mutate: $v = x_{r1} + 0.8(x_{r2} - x_{r3})$; clip to bounds.
4. Binomial crossover at 0.9; force one mutant coordinate.
5. Evaluate trial using mean development CV RMSE.
6. Accept trial when no worse; defer updates until generation end.

Stop after exactly 300 objective calls. No hidden polishing.

# Performance Metrics and Benchmarking

Same 204 holdout rows; tuned values are mean ± sample SD over five seeds.

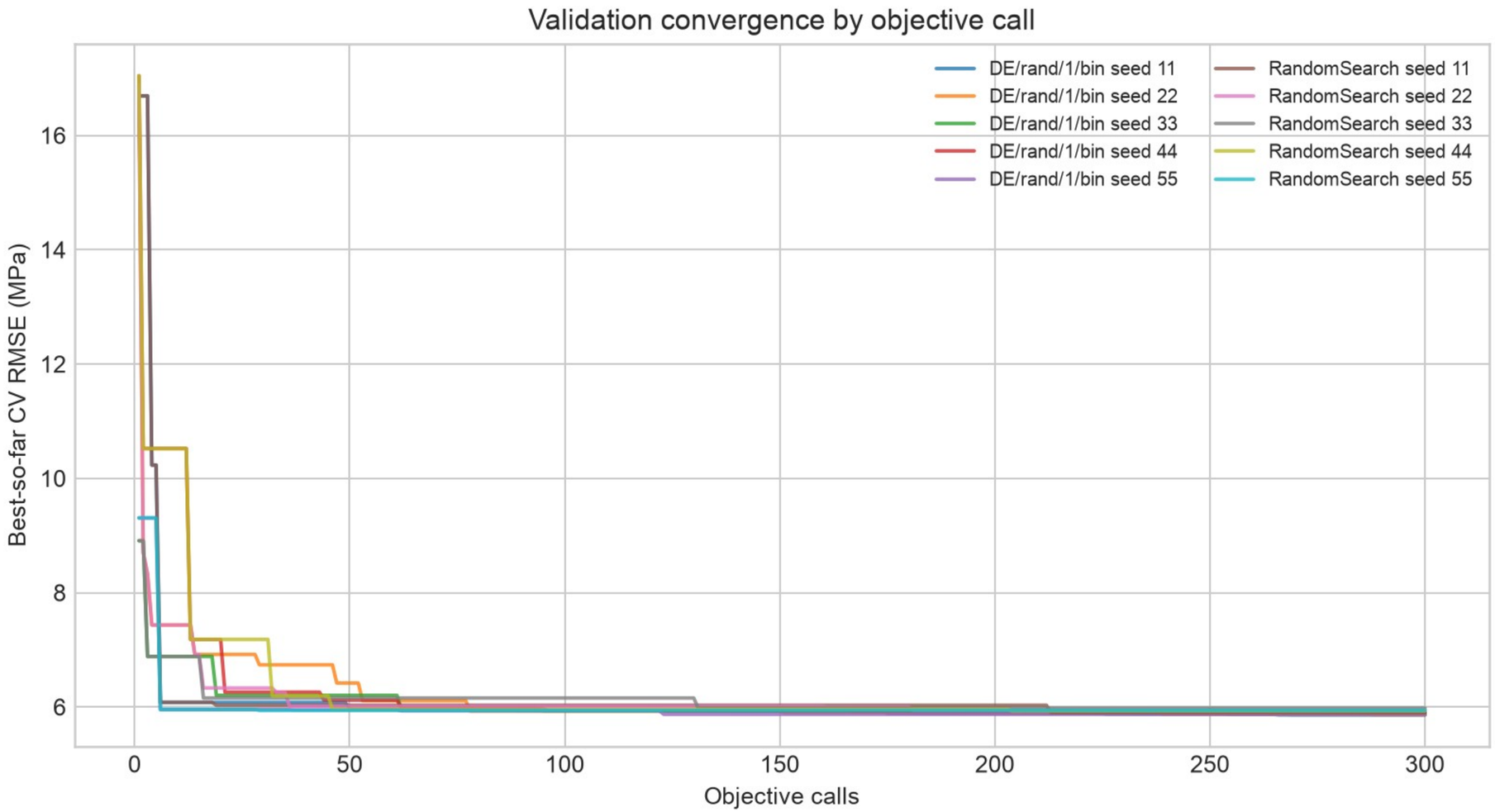
| Model | RMSE (MPa) | MAE (MPa) | $R^2$ |
|---|---|---|---|
| Default SVR | 9.9069 | 7.8854 | 0.5755 |
| Random Search–SVR | 6.2328 ± 0.1194 | 4.3502 ± 0.0253 | 0.8319 ± 0.0064 |
| **DE–SVR** | **6.2250 ± 0.1104** | **4.3143 ± 0.0526** | **0.8324 ± 0.0060** |

DE reduces RMSE by about 37.2% relative to default SVR.

DE and Random Search differ by only 0.0078 MPa (about 0.1%).

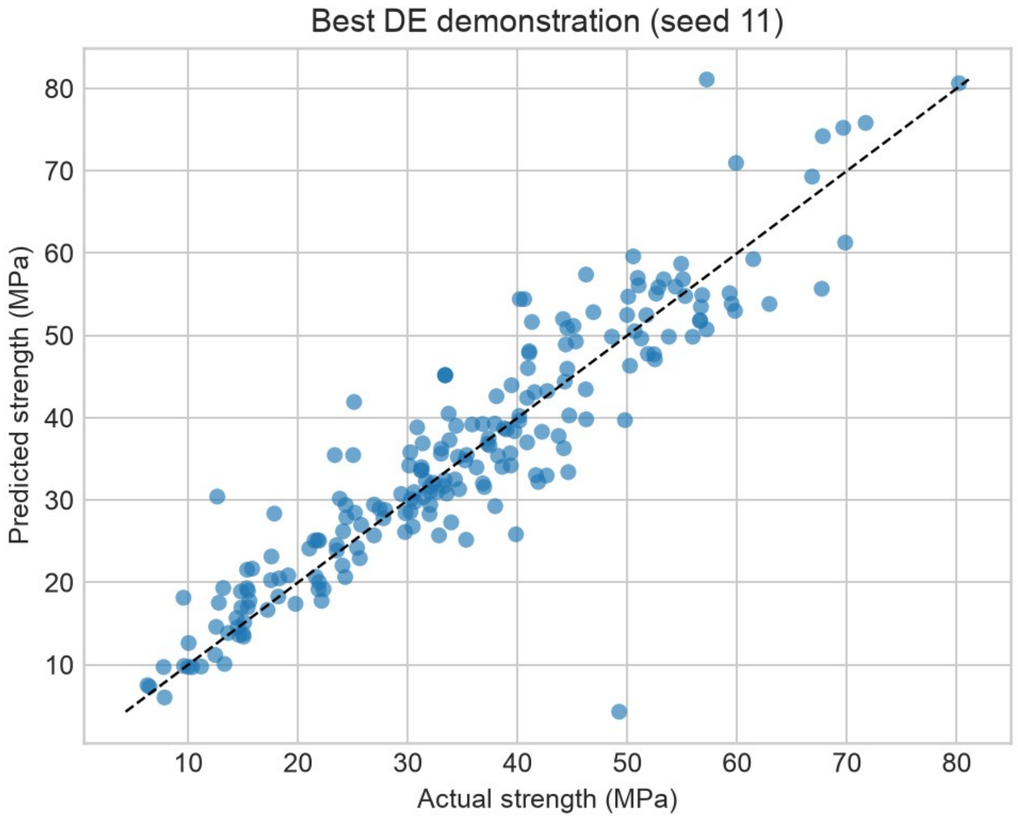
Conclusion: tuning helps; the two searches are essentially equivalent here.

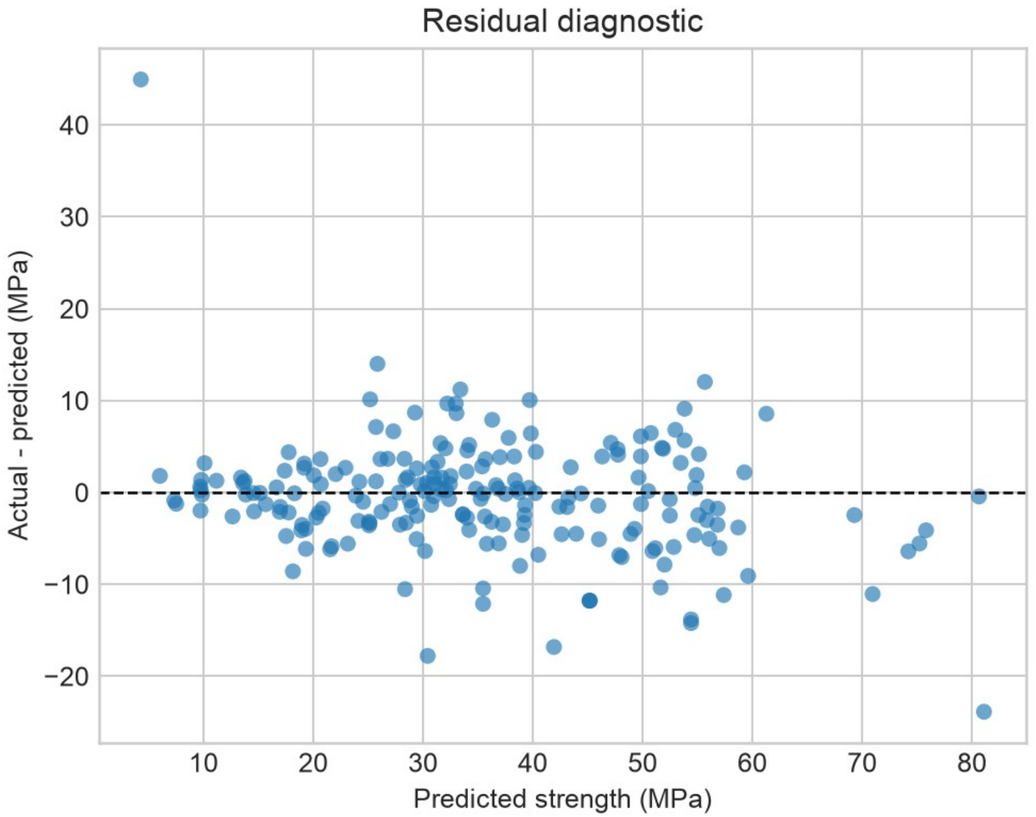
# Development Validation Convergence




Equal budgets: 300 calls per method per seed, including initialization.

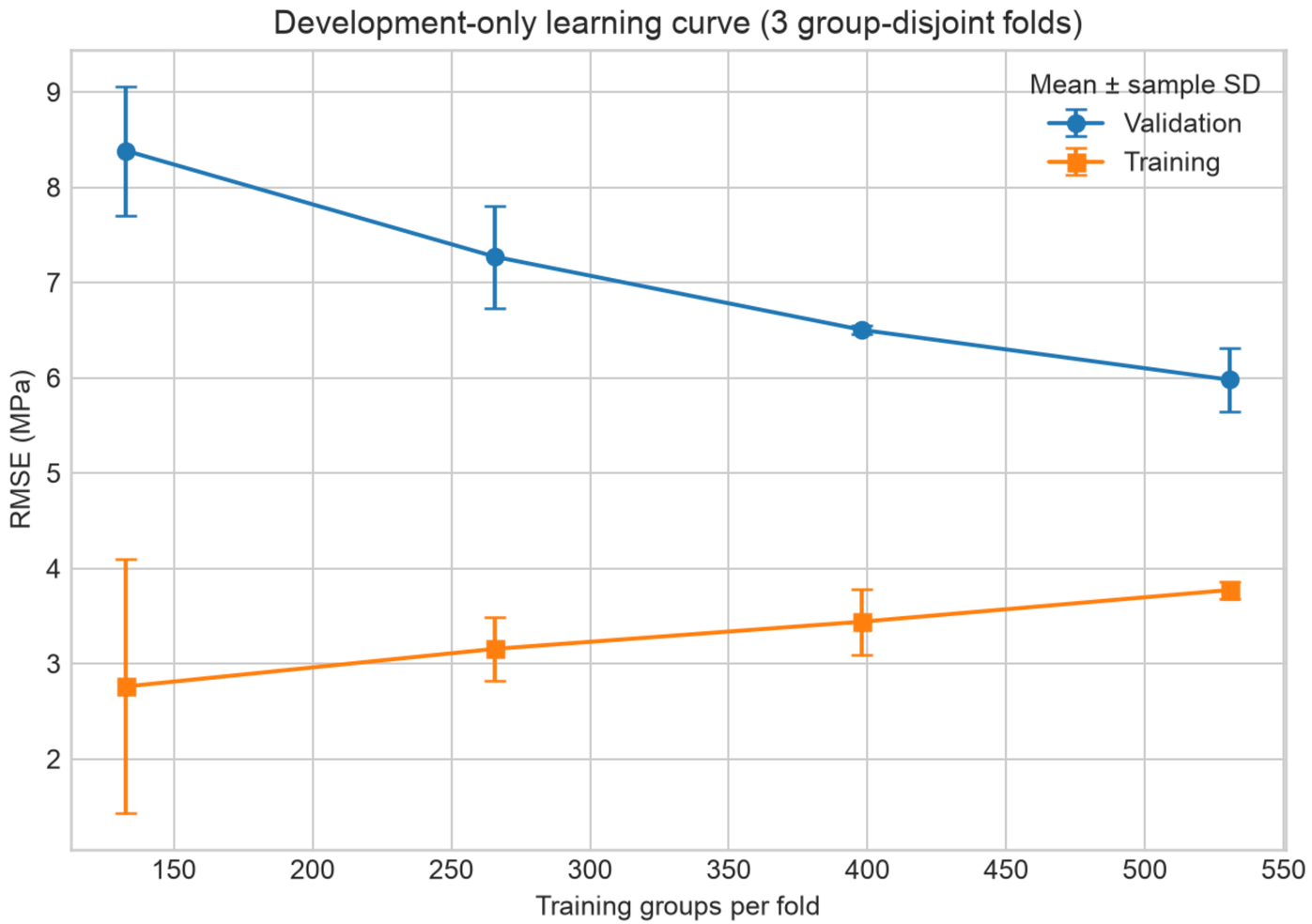
# Predictions and Residuals




Demonstration: DE seed 11, chosen by validation; holdout RMSE 6.3712 MPa.

# Learning-Curve Diagnostic




Three development-only folds; nested whole groups; fixed parameters; no retuning.

# Advantages

## What this approach offers

Explicit, tested DE operators for a three-parameter continuous search.

Both tuned methods substantially improve default SVR.

Group-aware splits and fold-local scaling reduce leakage risks.

Equal candidate budgets make the search comparison auditable.

Complete histories, predictions and locked dependencies support reproduction.

Forty-three tests and a saved evidence audit support technical closeout.

# Limitations

## What the evidence cannot establish

One public dataset and one fixed development / holdout partition.

Seed SD describes optimizer variability, not independent dataset uncertainty.

DE does not show meaningful superiority over equally budgeted Random Search.

Baseline and Random Search holdout evaluations preceded final DE evaluation.

No independent feature-importance or formal significance study.

Finite search cannot prove global optimality; no field deployment is validated.

# Future Scope

## Next experiments

Evaluate on external laboratory data using a frozen model.

Use repeated outer group splits or nested group cross-validation.

Test DE control settings systematically under matched budgets.

Study uncertainty and age-stratified errors with a declared protocol.

Add interpretation studies without using holdout feedback for tuning.

Assess material or environmental benefits only with suitable new evidence.

# References

## Selected sources

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[2] I.-C. Yeh, Cement and Concrete Research, 28(12), 1797–1808, 1998.

[3] R. Storn and K. Price, Journal of Global Optimization, 11, 341–359, 1997.

[4] J. Bergstra and Y. Bengio, JMLR, 13, 281–305, 2012.

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[6] scikit-learn documentation: SVR, StandardScaler and GroupKFold.

# THANK YOU

Both tuned methods improve default SVR.

DE and Random Search perform essentially equivalently.

github.com/R-V-Abhishek/concrete-strength-de-svr